

1_10xMouse_RA_Stellarscope

August 31, 2026

```
[25]: library(Seurat)
library(ggplot2)
library(ggpubr)
library(pheatmap)
library(dplyr)
library(RColorBrewer)
getwd()
dir.create("figures_10xMouse_RA")
dir.create("data_10xMouse_RA")
colorDict = c("Differentiating"="#88535A",
              "2CLC"="#EF8264",
              "Pluripotent"="#F2CC8F")

dataset_id <- "10xMouse_RA"
sample <- "LIF"
```

```
'/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'
```

```
Warning message in dir.create("figures_10xMouse_RA"):
"'figures_10xMouse_RA' already exists"
```

```
Warning message in dir.create("data_10xMouse_RA"):
"'data_10xMouse_RA' already exists"
```

```
[3]: dataset_id <- "10xMouse_RA"
sample <- "LIF"
path <- paste0("/mnt/TEresults2/snakemake_results/results/stellarscope_out/",
               dataset_id, "/", sample, "/pseudobulk/")

STAR_path <- "/mnt/TEresults/snakemake_results/results/STARoutdir/10xMouse_RA/
             LIF/best_Solo.out/Gene"
filteredBarcodes <- read.table(paste0(STAR_path, "/filtered/barcodes.tsv"))$V1
             # read barcodes selected by STARsolo
```

```

twoCLC_TEmatrix <- Seurat::ReadMtx(mtx = paste0(path, sample,
  ↪ "_pseudobulk-TE_counts.mtx"),
                                cells = paste0(path, sample,
  ↪ "_pseudobulk-barcodes.tsv"),
                                features = paste0(path, sample,
  ↪ "_pseudobulk-features.tsv"),
                                feature.column = 1) # read matrix
twoCLC_TEmatrix <-
  ↪ twoCLC_TEmatrix[setdiff(rownames(twoCLC_TEmatrix), "__no_feature"), filteredBarcodes]

nCells <- ncol(twoCLC_TEmatrix)
thrMinCells <- round(nCells * 0.02)
thrMinCells

# create Seurat object with shallow filtering of TEs expressed in at least 20
  ↪ cells and cells expressing at least 50 TEs
objTE_stellarscope <- Seurat::CreateSeuratObject(twoCLC_TEmatrix, project =
  ↪ "2CLC",
                                                min.cells = thrMinCells, min.features = 50)

#objTE_stellarscope <- objTE_allCB[,filteredBarcodes] # keep only filtered
  ↪ barcodes

objTE_stellarscope

```

31

Warning message:

"Feature names cannot have underscores ('_'), replacing with dashes ('-')"

An object of class Seurat

39250 features across 1542 samples within 1 assay

Active assay: RNA (39250 features, 0 variable features)

1 layer present: counts

```
[4]: summary(rowSums(objTE_stellarscope@assays$RNA$counts > 0))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
31	40	57	92	98	1527

```

[5]: annotation_stellarscope <- read.table("annotation/annotation_stellarscope.tsv")
  ↪ # generated with annotation_scripts/create_annotations_mouse.Rmd
head(annotation_stellarscope)

```

		chr	start	end	sw_score	strand	locusID	family	subfam
		<chr>	<int>	<int>	<int>	<chr>	<chr>	<chr>	<chr>
A data.frame: 6 × 10	1	chr1	3000001	3002128	12955	-	L1_Mus3_dup24	L1	L1-Mu
	2	chr1	3003153	3003994	1216	-	L1Md_F_dup2	L1	L1Md-
	3	chr1	3003994	3004054	234	-	L1_Mus3_dup25	L1	L1-Mu
	4	chr1	3004041	3004206	3685	+	L1_Rod_dup1	L1	L1-Ro
	5	chr1	3004271	3005001	3685	+	L1_Rod_dup2	L1	L1-Ro
	6	chr1	3005002	3005439	1280	+	L1_Rod_dup3	L1	L1-Ro

```
[6]: # Remove some classes of TEs

classesToExclude <- c("Other", "Satellite", "Unknown", "RNA")

stellarscopeTEs <- Features(objTE_stellarscope)

TEsToKeep <- annotation_stellarscope[! annotation_stellarscope$class %in%
  ↪classesToExclude, ]$stellarscopeID

length(setdiff(stellarscopeTEs, TEsToKeep))/ length(stellarscopeTEs) * 100 #
  ↪percentage removed

objTE_stellarscope <- objTE_stellarscope[intersect(stellarscopeTEs, TEsToKeep),]
```

1.36815286624204

```
[7]: feature_metadata <- annotation_stellarscope[match(Features(objTE_stellarscope),
  ↪annotation_stellarscope$stellarscopeID),]

head(feature_metadata)
```

		chr	start	end	sw_score	strand	locusID	family
		<chr>	<int>	<int>	<int>	<chr>	<chr>	<chr>
A data.frame: 6 × 10	64	chr1	3057320	3059492	18188	+	RLTR45-int_dup6	ERVK
	89	chr1	3076927	3077419	4306	-	MT2_Mm_dup2	ERVL
	176	chr1	3143093	3143442	2310	-	IAPEY2_LTR	ERVK
	206	chr1	3168309	3168674	657	+	MLT1A_dup2	ERVL-MaLH
	211	chr1	3171937	3175248	16392	+	L1_Mus4_dup3	L1
	213	chr1	3175517	3175992	4198	-	MT2_Mm_dup3	ERVL

```
[8]: options(repr.plot.width=7, repr.plot.height=5)

objTE_stellarscope@meta.data$nCount_TE <- objTE_stellarscope@meta.
  ↪data$nCount_RNA
objTE_stellarscope@meta.data$nFeature_TE <- objTE_stellarscope@meta.
  ↪data$nFeature_RNA
# Visualize QC metrics as a violin plot
VlnPlot(objTE_stellarscope, features = c("nCount_TE"), ncol = 1,
```

```

      cols = colorDict, pt.size = 0) + theme(text=element_text(size=17))

ggsave("figures_10xMouse_RA/nCountTE_violin_stellarscope.png",
      ↪device='png',dpi=600)
ggsave("figures_10xMouse_RA/nCountTE_violin_stellarscope.pdf", device='pdf')

VlnPlot(objTE_stellarscope, features = c("nFeature_TE"), ncol = 1,
      cols = colorDict, pt.size = 0) + theme(text=element_text(size=17))
ggsave("figures_10xMouse_RA/nFeatureTE_violin_stellarscope.png",
      ↪device='png',dpi=600)
ggsave("figures_10xMouse_RA/nFeatureTE_violin_stellarscope.pdf", device='pdf')

```

Warning message:

"Default search for "data" layer in "RNA" assay yielded no results; utilizing "counts" layer instead."

Warning message:

"The `slot` argument of `FetchData()` is deprecated as of SeuratObject 5.0.0.

Please use the `layer` argument instead.

The deprecated feature was likely used in the [Seurat](#) package.

Please report the issue at

[<https://github.com/satijalab/seurat/issues>.](https://github.com/satijalab/seurat/issues)"

Warning message:

"`PackageCheck()` was deprecated in SeuratObject 5.0.0.

Please use `rlang::check\_installed()` instead.

The deprecated feature was likely used in the [Seurat](#) package.

Please report the issue at

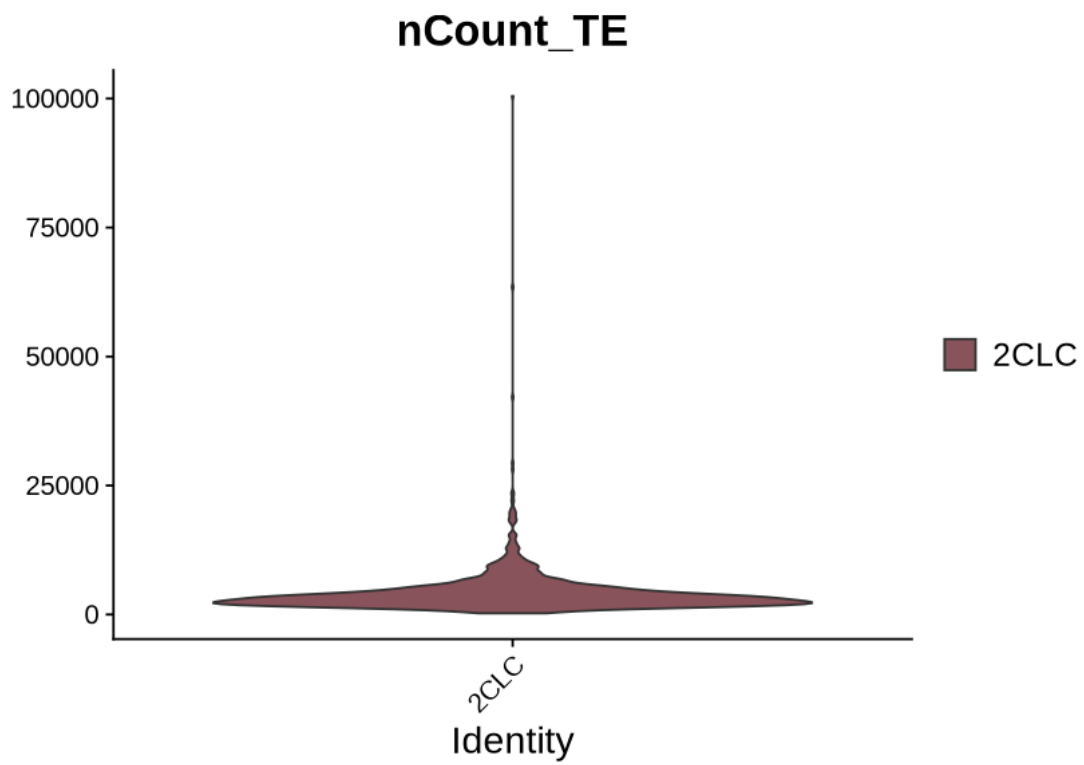
[<https://github.com/satijalab/seurat/issues>.](https://github.com/satijalab/seurat/issues)"

Saving 7 x 7 in image

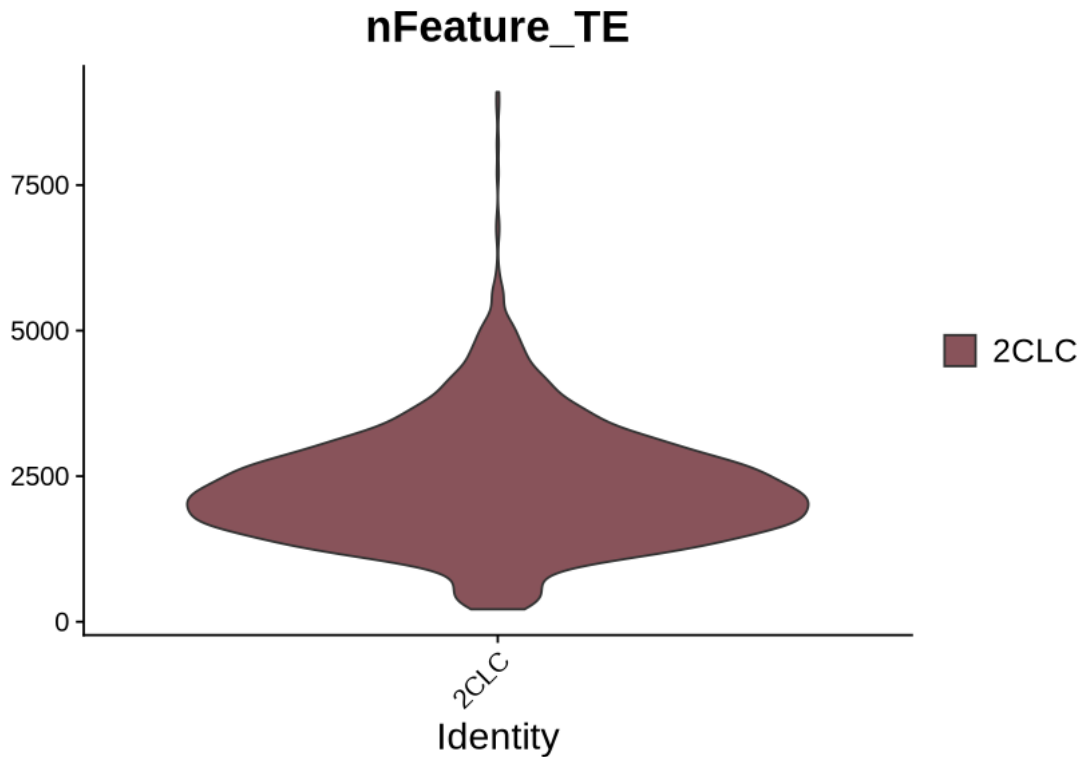
Saving 7 x 7 in image

Warning message:

"Default search for "data" layer in "RNA" assay yielded no results; utilizing "counts" layer instead."



Saving 7 x 7 in image
Saving 7 x 7 in image



```
[9]: objTE_stellarscope <- JoinLayers(objTE_stellarscope)

objTE_stellarscope <- NormalizeData(objTE_stellarscope, normalization.method = "LogNormalize",
  scale.factor = 10000)

objTE_stellarscope <- FindVariableFeatures(objTE_stellarscope, selection.method = "vst",
  nfeatures = 4000)

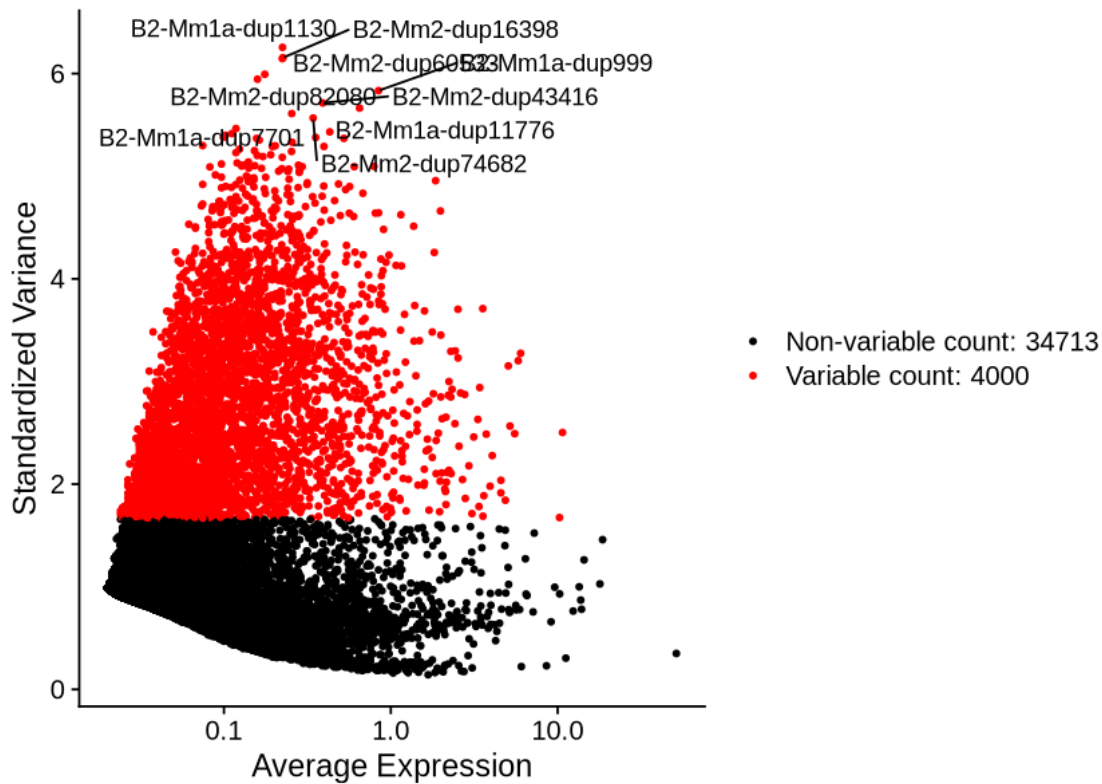
# Identify the 10 most highly variable genes
top10 <- head(VariableFeatures(objTE_stellarscope), 10)

# plot variable features with and without labels
plot1 <- VariableFeaturePlot(objTE_stellarscope)
plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
plot2
```

Normalizing layer: counts

Finding variable features for layer counts

When using repel, set xnudge and ynudge to 0 for optimal results



```
[10]: objTE_stellarscope
```

```
An object of class Seurat
38713 features across 1542 samples within 1 assay
Active assay: RNA (38713 features, 4000 variable features)
2 layers present: counts, data
```

```
[11]: gc()
all.genes <- rownames(objTE_stellarscope)
objTE_stellarscope <- ScaleData(objTE_stellarscope) # on hugs
```

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2 × 6 of type dbl	Ncells	14752799	787.9	43114697	2302.6	28463238	1520.2
	Vcells	107918258	823.4	198423975	1513.9	197809748	1509.2

Centering and scaling data matrix

```
[12]: grep("MERVL", all.genes, value = T)[1:20]
```

```
1. 'MERVL-int-dup2' 2. 'MERVL-int-dup5' 3. 'MERVL-int-dup8' 4. 'MERVL-int-dup10'
5. 'MERVL-int-dup13' 6. 'MERVL-int-dup19' 7. 'MERVL-int-dup23' 8. 'MERVL-int-dup29'
9. 'MERVL-int-dup30' 10. 'MERVL-int-dup33' 11. 'MERVL-int-dup35' 12. 'MERVL-int-dup50'
```

13. 'MERVL-int-dup51' 14. 'MERVL-int-dup58' 15. 'MERVL-int-dup63' 16. 'MERVL-int-dup67'
 17. 'MERVL-int-dup75' 18. 'MERVL-int-dup78' 19. 'MERVL-int-dup81' 20. 'MERVL-int-dup84'

```
[13]: objTE_stellarscope <- RunPCA(objTE_stellarscope, features =  

  ↪VariableFeatures(object = objTE_stellarscope))  
  

  DimPlot(objTE_stellarscope, reduction = "pca") + NoLegend()  
  

  ElbowPlot(objTE_stellarscope)
```

PC_ 1

Positive: ERVB3-1-I-MM-int-dup14, B2-Mm2-dup63899, B2-Mm2-dup82766,
 RLTR45-dup757, B1-Mus1-dup76311, RMER16-Mm-dup425, B2-Mm2-dup55156, RLTR45-int-
 dup1196, B2-Mm2-dup26429, B2-Mm2-dup60400

B1-Mus2-dup44255, B1-Mus2-dup6743, B1-Mur1-dup17427,
 B1-Mus1-dup50269, ID4-dup13019, B1-Mus2-dup19537, B2-Mm2-dup37101,
 RLTR9D2-dup135, B1-Mus1-dup71993, B2-Mm2-dup9168

B1-Mm-dup9681, B1-Mm-dup41356, L1MB8-dup5667, B2-Mm1a-dup9981,
 ETnERV3-int-dup124, B2-Mm2-dup49684, B2-Mm1t-dup2058, B2-Mm2-dup59014,
 RMER16-int-dup2575, RLTR45-dup569

Negative: B2-Mm1a-dup14558, B2-Mm1a-dup14535, B2-Mm1a-dup15576,
 B2-Mm1a-dup12142, B2-Mm1t-dup14643, B2-Mm1a-dup10686, B2-Mm1a-dup7379,
 B2-Mm1a-dup13490, B2-Mm1a-dup11294, B2-Mm1a-dup17482

B2-Mm2-dup14391, B2-Mm1t-dup17879, B2-Mm1t-dup2101, B2-Mm1a-dup6847,
 B2-Mm1a-dup15696, B2-Mm1a-dup7813, B2-Mm2-dup38946, B2-Mm1a-dup2100,
 B2-Mm1a-dup5893, B2-Mm2-dup77593

B2-Mm1t-dup18260, B2-Mm1a-dup8415, B2-Mm1a-dup3101, B2-Mm1t-dup3367,
 B2-Mm1a-dup15898, B2-Mm2-dup57274, B2-Mm1t-dup16798, B2-Mm1t-dup15736,
 B2-Mm1a-dup12691, B2-Mm1a-dup12318

PC_ 2

Positive: B2-Mm1a-dup3997, B2-Mm1a-dup4894, B2-Mm1a-dup3798, B2-Mm2-dup89012,
 L1MB4-dup2499, B2-Mm1a-dup12769, B2-Mm2-dup16477, B2-Mm1t-dup11152,
 B2-Mm1a-dup5423, L1MB8-dup5667

B2-Mm1t-dup8107, B2-Mm2-dup49317, MMERVK10C-int-dup2240,
 B2-Mm2-dup60347, B2-Mm1a-dup2798, B2-Mm2-dup24437, B2-Mm1t-dup2762,
 L1MB4-dup2500, PB1D7-dup17870, B2-Mm1a-dup17758

B2-Mm1a-dup13487, FordPrefect-dup264, RMER16-Mm-dup425,
 B2-Mm1a-dup14850, B2-Mm2-dup15962, B1-Mus1-dup88862, B3-dup74176,
 B2-Mm2-dup17845, B2-Mm1t-dup16889, B2-Mm1t-dup11485

Negative: L1Md-F3-dup8859, L1Md-F2-dup38432, B3A-dup15576, B3-dup23645,
 MT2B1-dup1331, IAPez-int-dup4789, MTC-dup25430, Lx5-dup5180, L1-Mur2-dup20290,
 Lx2A-dup1335

Lx3A-dup8033, ORR1A1-dup3177, Lx2B-dup5998, MT2-Mm-dup1923,
 B1-Mus1-dup88035, Lx3A-dup1428, MLTR31D-MM-dup144, B1-Mus1-dup14857, Lx3-Mus-
 dup264, MER21-int-dup86

RSINE1-dup37336, HAL1-dup1240, MT2-Mm-dup1199, L1-Mur3-dup6967, MTA-
 Mm-dup12737, Lx2B2-dup8361, LTRIS-Mus-dup747, B3A-dup15577, MER33-dup385,
 MT2B1-dup1304

PC_ 3

Positive: B3-dup116655, B2-Mm1t-dup21713, B2-Mm2-dup79569, B2-Mm1a-dup6903, B2-Mm1t-dup19816, B2-Mm2-dup61969, B2-Mm1a-dup3614, B2-Mm2-dup41938, B2-Mm2-dup18650, B2-Mm2-dup67657

B2-Mm1a-dup14678, B2-Mm2-dup23671, B2-Mm2-dup56519, B2-Mm2-dup50542, B2-Mm2-dup69910, B2-Mm2-dup81484, B2-Mm1t-dup8905, B2-Mm1t-dup20953, B2-Mm1a-dup14225, B2-Mm2-dup62045

B2-Mm1a-dup7833, B3A-dup70742, B2-Mm2-dup55513, B1-Mus1-dup8091, B2-Mm1t-dup7646, B2-Mm1t-dup14131, B2-Mm2-dup6480, B2-Mm1t-dup7915, B2-Mm1a-dup10095, B4-dup58342

Negative: MusHAL1-dup3315, B2-Mm1t-dup287, B2-Mm1t-dup18617, B2-Mm2-dup80011, MT2B2-dup1441, MT2B1-dup4621, B2-Mm1t-dup5959, B2-Mm2-dup49674, B2-Mm1a-dup14762, B2-Mm1a-dup7388

B2-Mm1a-dup1575, B2-Mm2-dup16398, B2-Mm1t-dup3581, MMERVK10C-int-dup383, B2-Mm2-dup74765, B2-Mm1t-dup6383, B2-Mm2-dup1305, B2-Mm1t-dup13858, B2-Mm2-dup26151, B2-Mm1t-dup13286

MTA-Mm-dup12737, B2-Mm2-dup73383, B2-Mm2-dup32792, B2-Mm2-dup25868, MMERVK10C-int-dup929, B2-Mm1a-dup17524, B2-Mm1t-dup21209, B1-Mus1-dup22730, ORR1B1-dup13239, B2-Mm1a-dup15361

PC_ 4

Positive: B2-Mm2-dup14391, B1-Mus1-dup37892, B2-Mm1t-dup2238, B2-Mm1a-dup14558, B2-Mm1a-dup4522, B1-Mus2-dup11738, B2-Mm1a-dup14535, B2-Mm2-dup5176, B2-Mm1t-dup5626, B2-Mm1a-dup16034

MMERGLN-int-dup17, B2-Mm2-dup14405, B2-Mm1t-dup15753, B2-Mm1a-dup18091, B2-Mm1a-dup17531, B2-Mm2-dup18987, B2-Mm1a-dup14649, B2-Mm1t-dup15734, B2-Mm1t-dup8107, B2-Mm1a-dup15239

B2-Mm1a-dup16639, B2-Mm1t-dup7554, B2-Mm1t-dup18897, B2-Mm1t-dup15475, B2-Mm1a-dup6070, B1-Mus2-dup69176, B2-Mm1a-dup11779, B2-Mm1a-dup7770, B1-Mus2-dup32414, B1-Mus2-dup33145

Negative: B2-Mm1a-dup15361, B2-Mm2-dup57274, B2-Mm1t-dup4719, B2-Mm1a-dup14762, B2-Mm1t-dup5959, B2-Mm1a-dup1575, B2-Mm1t-dup5807, B2-Mm2-dup49674, B2-Mm2-dup1305, B2-Mm1t-dup287

B2-Mm1t-dup13286, B2-Mm2-dup80011, B2-Mm1a-dup7388, B2-Mm2-dup39326, B2-Mm1a-dup8099, MT2B1-dup6057, B2-Mm2-dup74765, MusHAL1-dup3315, B2-Mm1a-dup10730, B2-Mm1t-dup18617

B2-Mm1t-dup1766, B1-Mus1-dup22730, B2-Mm2-dup44726, B2-Mm1a-dup3248, MT2B1-dup4621, B2-Mm1t-dup10631, B2-Mm1t-dup1041, B2-Mm2-dup34599, B1-Mm-dup21910, B2-Mm2-dup8058

PC_ 5

Positive: Lx3-Mus-dup3995, Lx3-Mus-dup4580, B2-Mm1a-dup6716, B2-Mm1t-dup18248, B2-Mm1a-dup15900, MMERVK10C-int-dup929, B2-Mm1a-dup2155, B2-Mm2-dup12871, B2-Mm1t-dup5698, B2-Mm1a-dup13598

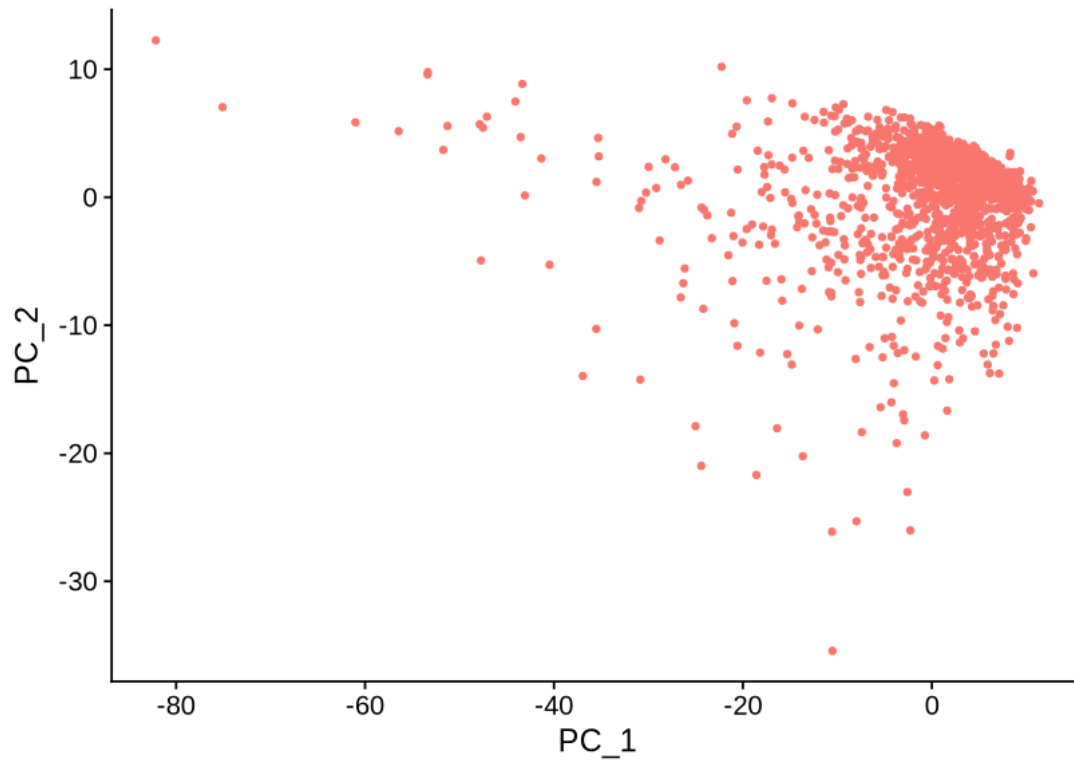
B2-Mm1t-dup3616, B2-Mm2-dup713, B2-Mm2-dup79434, B2-Mm1a-dup25, B2-Mm1a-dup2453, B2-Mm2-dup82756, B2-Mm1t-dup16966, B2-Mm2-dup33763, B2-Mm1a-dup11065, MMERVK10C-int-dup922

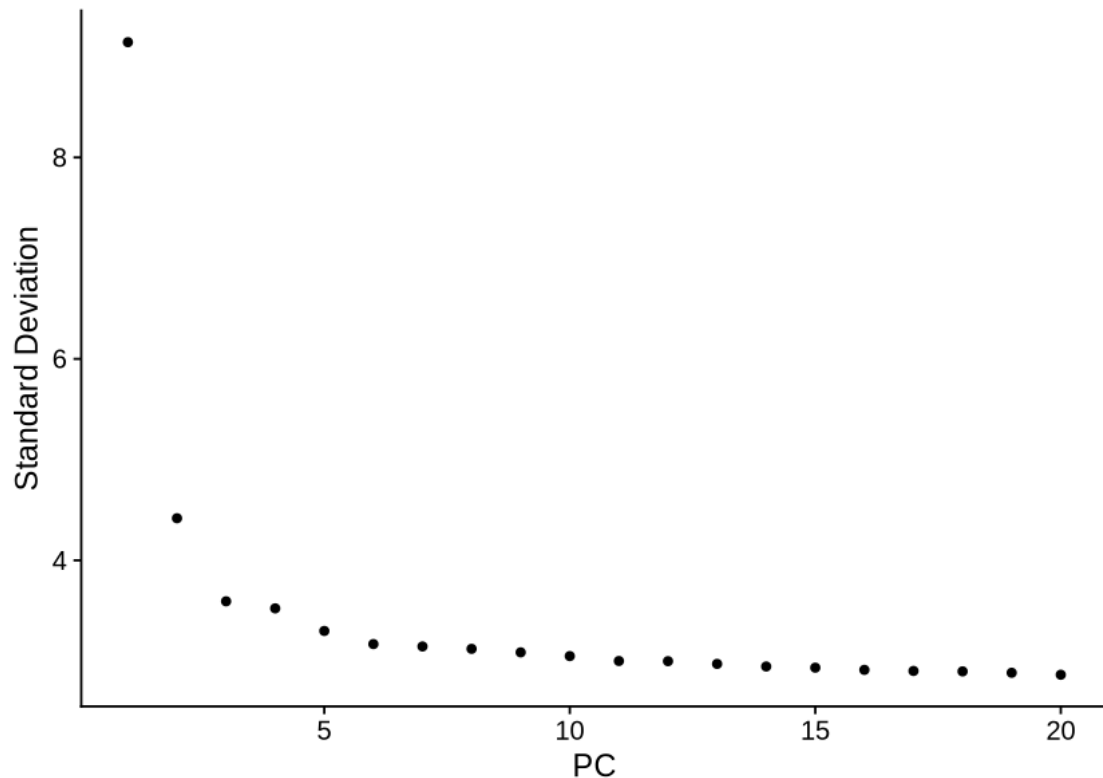
B2-Mm1a-dup8415, B2-Mm1t-dup11314, B2-Mm1a-dup2815, B2-Mm1a-dup7362, B2-Mm2-dup59584, MYSERV-int-dup209, B2-Mm2-dup50672, B2-Mm1t-dup15680, B2-Mm1a-dup14678, B2-Mm1a-dup12238

Negative: B3-dup23645, Lx2B-dup5998, Lx2A-dup1335, B2-Mm2-dup1305, L1-Mur2-dup20290, RSINE1-dup37336, B1-Mus2-dup39024, ID-B1-dup50226, MTC-dup25430, B1-Mus2-dup69523

B3A-dup15576, B2-Mm2-dup31937, B1-Mm-dup3137, MT2-Mm-dup1923, B1-Mm-dup33050, B1-Mus2-dup64613, B1-Mm-dup27019, B1-Mm-dup21910, Lx3A-dup1428, B1-Mus1-dup25823

B1-Mus1-dup14857, B2-Mm2-dup23671, B2-Mm1t-dup18617, B1-Mus2-dup32414, RLTR45-int-dup1196, L1MB4-dup2499, B1-Mm-dup23509, B3A-dup15577, ETnERV3-int-dup126, B1-Mus1-dup88035





```
[4]: objTE_stellarscope <- FindNeighbors(objTE_stellarscope, dims = 1:15, k.param = 20)
      objTE_stellarscope <- FindClusters(objTE_stellarscope, resolution = 1, algorithm=4)
      objTE_stellarscope <- RunUMAP(objTE_stellarscope, dims = 1:15)
      DimPlot(objTE_stellarscope, reduction = "umap")
```

Computing nearest neighbor graph

Computing SNN

```
Warning message in RunLeiden(object = object, method = method, partition.type =
"RBConfigurationVertexPartition", :
"`random.seed` must be greater than 0 for leiden clustering, resetting
`random.seed` to 1."
Warning message:
"The default method for RunUMAP has changed from calling Python UMAP via
reticulate to the R-native UWOT using the cosine metric
To use Python UMAP via reticulate, set umap.method to 'umap-learn' and metric to
'correlation'
This message will be shown once per session"
```


*
*
*
*
*
*
*
*
*
*
*
*
*
*
*
*
|

00:12:37 Writing NN index file to temp file
/mnt/TEdataStorage_2T/tmp/RtmpUqXrXg/file3e38295e4862a0

00:12:37 Searching Annoy index using 1 thread, search_k = 3000

00:12:37 Annoy recall = 100%

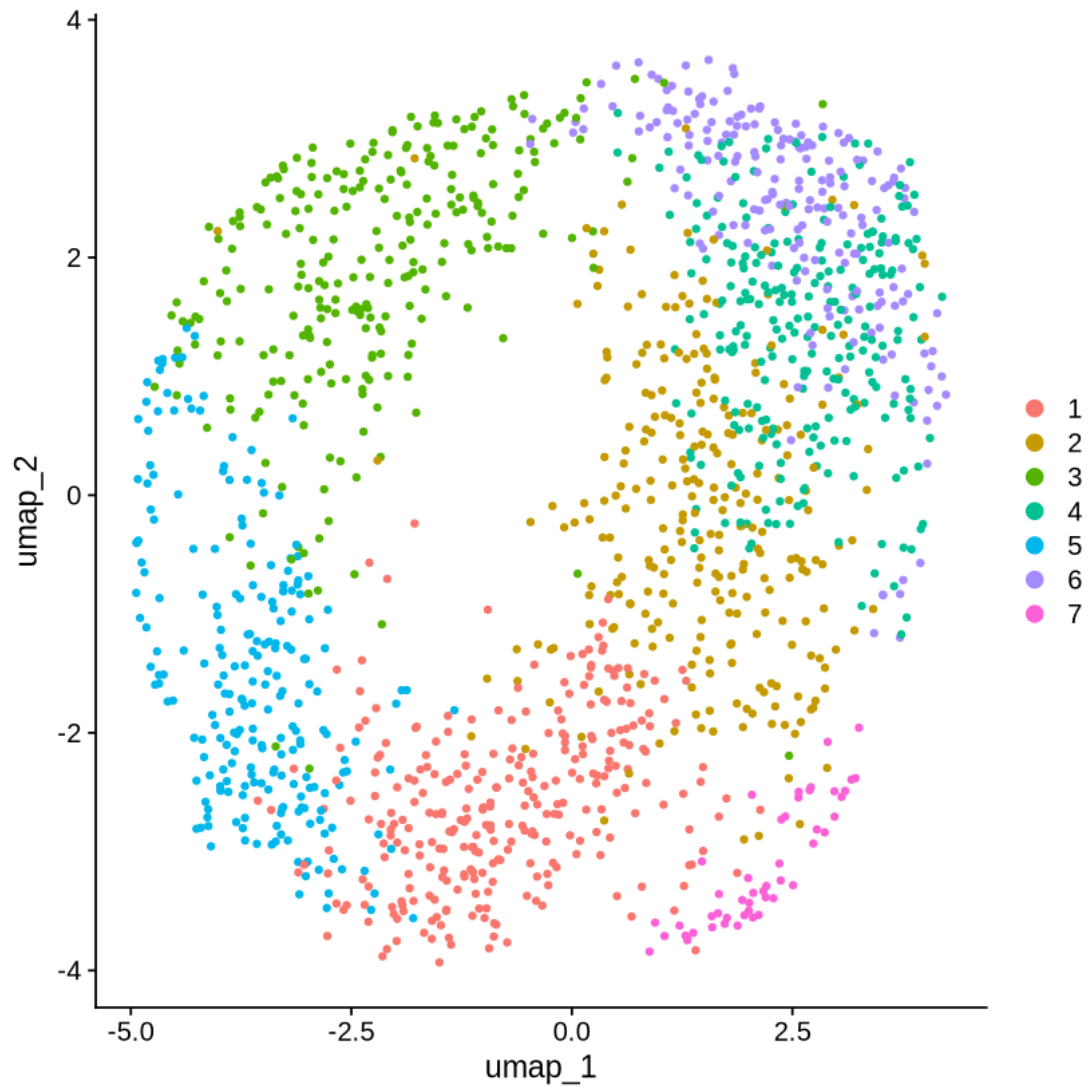
00:12:38 Commencing smooth kNN distance calibration using 1 thread
with target n_neighbors = 30

00:12:41 Initializing from normalized Laplacian + noise (using RSpectra)

00:12:41 Commencing optimization for 500 epochs, with 67976 positive edges

00:12:41 Using rng type: pcg

00:12:44 Optimization finished



```
[5]: FeaturePlot(objTE_stellarscope, reduction = "umap", features = "nCount_TE", pt.
      ↪size = 0.5) +
      theme_void() +
      theme(text=element_text(size=20))
```

Warning message:

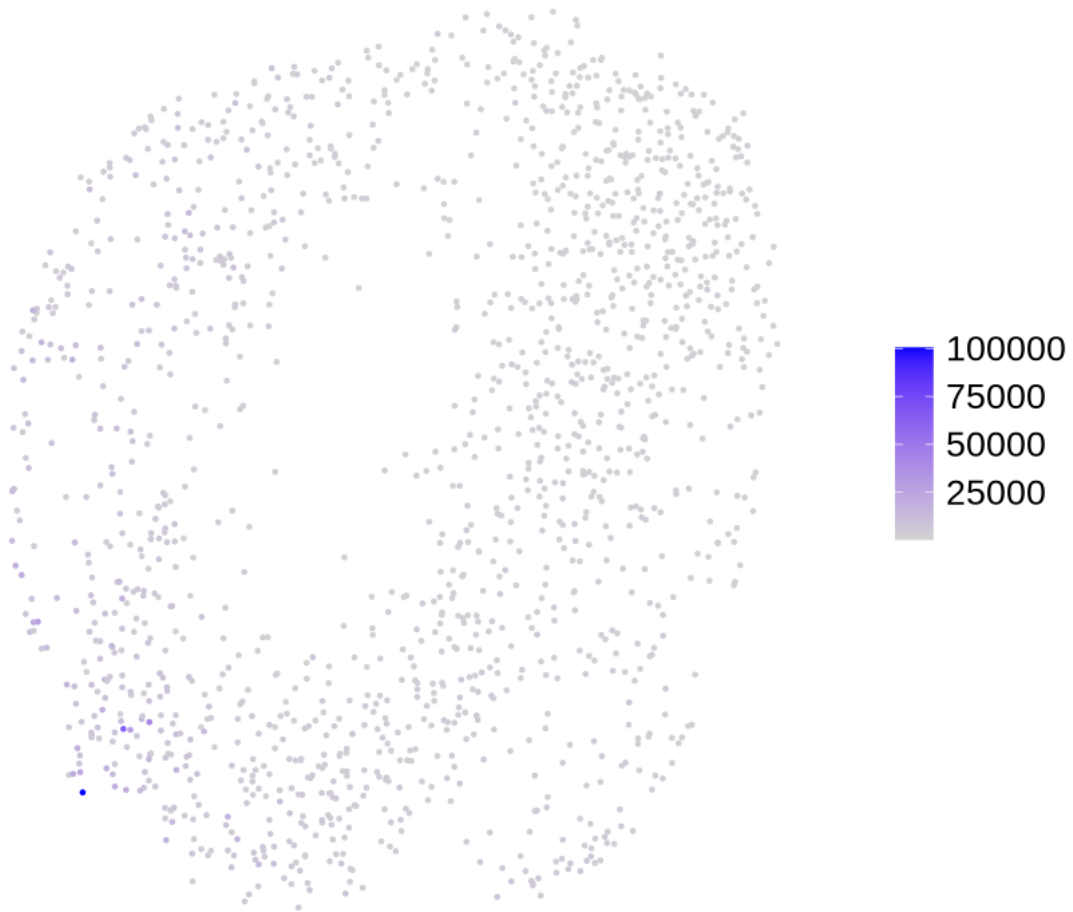
"The `slot` argument of `FetchData()` is deprecated as of SeuratObject 5.0.0.

Please use the `layer` argument instead.

The deprecated feature was likely used in the [Seurat](#) package.

Please report the issue at
<https://github.com/satijalab/seurat/issues>."

nCount_TE

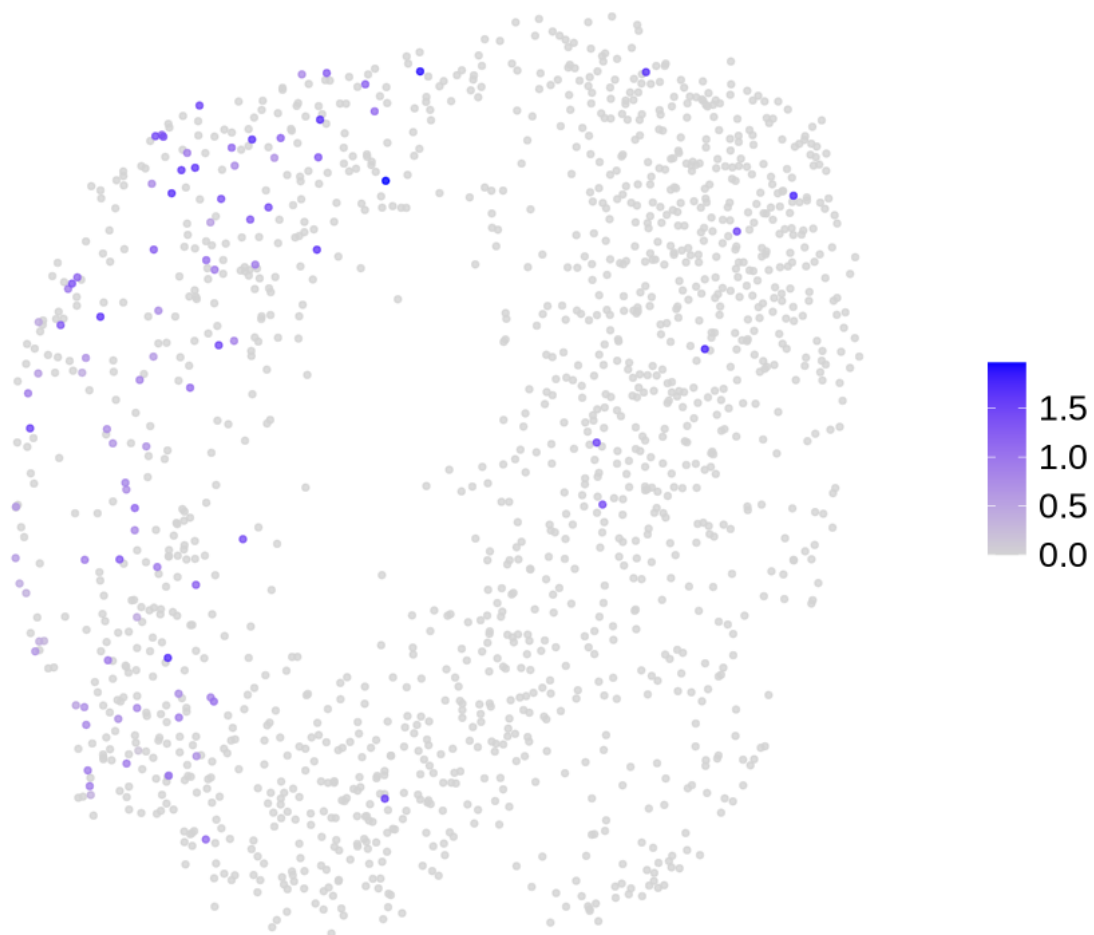


```
[6]: FeaturePlot(objTE_stellarscope, features =  
      ↪c('MERVL-int-dup60','MERVL-int-dup5'), pt.size=1,alpha = 0.8,order = TRUE )  
      ↪&  
      theme_void() &  
      theme(text=element_text(size=20))
```

Warning message:

"The following requested variables were not found: MERVL-int-dup60"

MERVL-int-dup5



```
[17]: #saveRDS(objTE_stellarscope, paste0("data_", dataset_id, "/stellarscope_",  
↪dataset_id, "_seuratObj_251125.RDS"))
```